



Systemic Rheumatologic Diseases

Principles and Practices of
Osteopathic Medicine 1

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**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Session Objectives

At the conclusion of the session/course, the student should be able to:

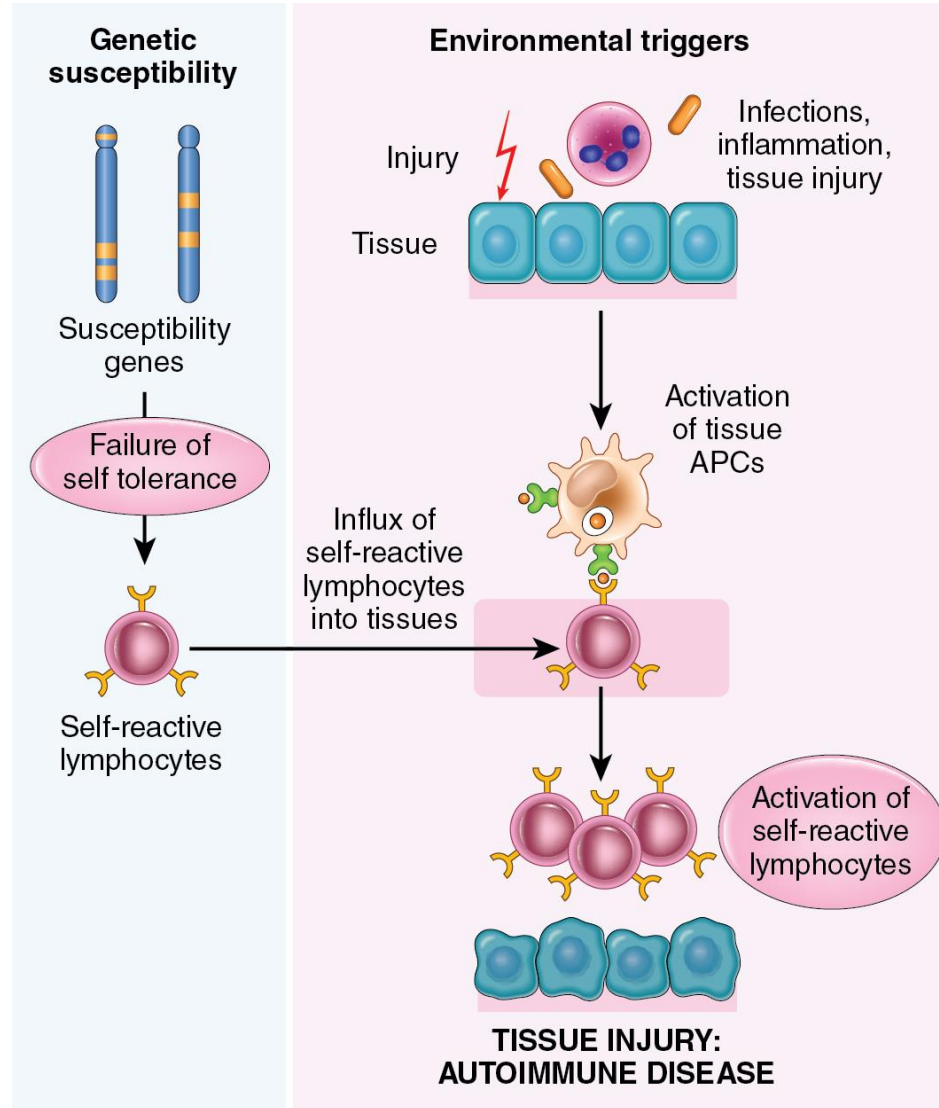
- Describe the presentation and diagnostic criteria used to diagnose Systemic Lupus Erythematosus (SLE).
- Describe the different uses of ANA testing.
- Describe the presentation of Systemic Sclerosis (Scleroderma)
- Describe the presentation of Sjorgen Syndrome

Autoimmunity Reminder

- Genetic susceptibility
- Breakdown of self tolerance
- Environmental triggers
- Tissue damage
- Activation of self-reactive lymphocytes
- Further Tissue damage

Why

How



Autoimmunity Considerations

- Considerations:
 - Chronic, relapses, remissions, progressive, systemic...
- Nature of underlying immune Response
- Hypersensitivity Reactions:
 - II: antibodies bind to cell surface causing destruction
 - III: Immune complex- antigen- antibody - complement

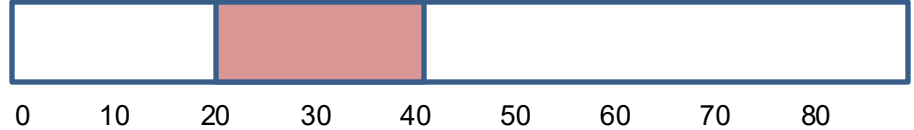
SLE: Epidemiology

Who



Sex:
9:1 F:M in
age 17-55
group

Genetic assoc.



Onset at 20-40 years



Race/Ethnicity

Most Common/severe in African,
Hispanic and Asian populations

Systemic Lupus Erythematosus (SLE)

- Multiple organ/system disorder
 - Effects mostly from Ab-Ag complex **deposition** in the capillaries of organs (*Hypersensitivity type III*)
- Autoimmune, inflammatory disorder
 - Auto-Ab to nuclear antigens
- Remission-relapse cycle
- May be mild and self-limited or can progress to life threatening end organ damage

SLE Pathogenesis

- Genetic Predisposition
 - B and T cells specific for self nuclear antigens
 - Defective tolerance
 - Defective clearance of apoptotic bodies
- Environmental Factors:
 - UV light
 - Drugs: hydralazine, procainamide, D-penicillamine.
- Harm: immune complex deposition and binding of antibodies..

Clinical pathogenesis of SLE

PREDISPOSING FACTORS

GENES

High Hazard Ratios (≥ 6);

Deficiencies of C1q, C2, C4 (rare)
TREX1 mutations affecting DNA degradation (rare)

Affecting Ag presentation or persistence, e.g., phagocytosis of immune complexes

HLA-DRB1 (*1501, *0301), DR3, DQA2
CR2, FCGR2A/B

Enhance Innate Immunity, including production of IFNs

TNFAIP3, IRF5/TNPO3, IRF7/PHRF1, ITGAM, ICAMs

Alter Adaptive Immunity B and/or T Cell Signaling

BANK1, STAT4, MSHS, IZKF3, TCF7

GENES FOR LUPUS NEPHRITIS

HLA-DR3, STAT4, APOL1 (African Americans),
FCGR3A, ITGAM, IRF5, IRF7, TNFSF4 (Ox40L), DNASE1

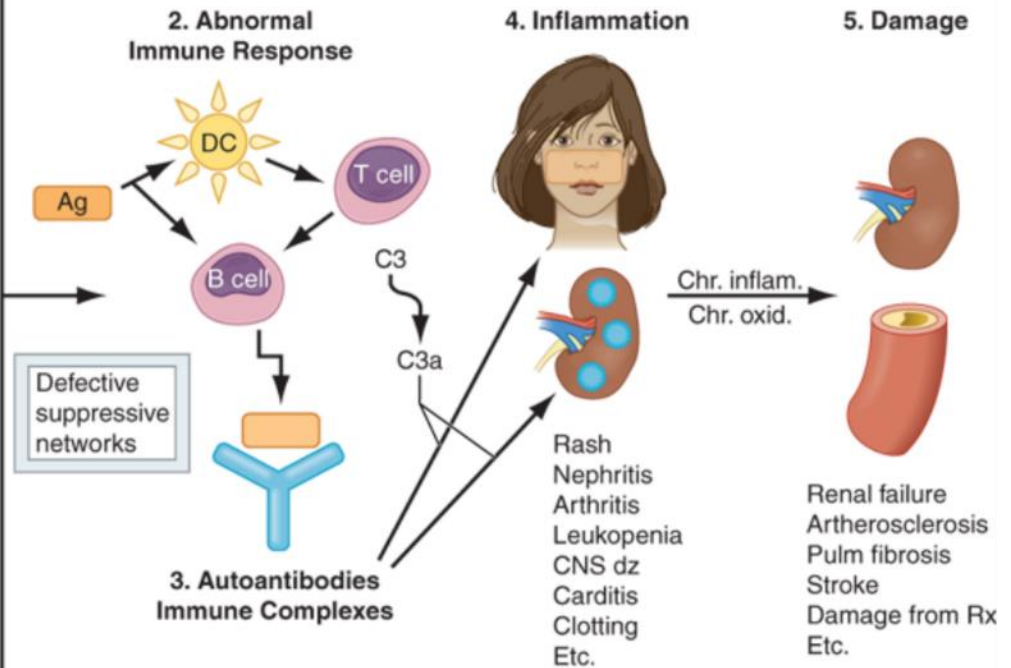
ENVIRONMENT/MICROENVIRONMENT

Ultraviolet Light, Smoking, Crystalline
Silica, ?EBV infection
Femaleness

EPIGENETICS

Hypomethylation of DNA: In CD4+T, B and monocytes
Some affect IFN production
Histone modifications: Some increase expression
of predisposing genes and/or IFN production
MicroRNA affecting gene expression

Mir-21, -146A, -155, -569, -30A, Let-7a



Citation: Systemic Lupus Erythematosus, Jameson J, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J. *Harrison's Principles of Internal Medicine*, 20e; 2018. Available at: <https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=192284871> Accessed: January 10, 2019

Presentations of SLE

- Constitutional: fever
- Dermatologic: Malar or discoid rashes
- Cardiac/Pulmonary: pleuritis, pericarditis
- Hematologic: cytopenias
- Musculoskeletal: nonerosive arthritis
- Neurologic: seizures or psychosis
- Renal: proteinuria or casts, nephritis

Most common presentation of SLE

- Malar Rash
 - AKA Butterfly Rash
 - Resolves without scar

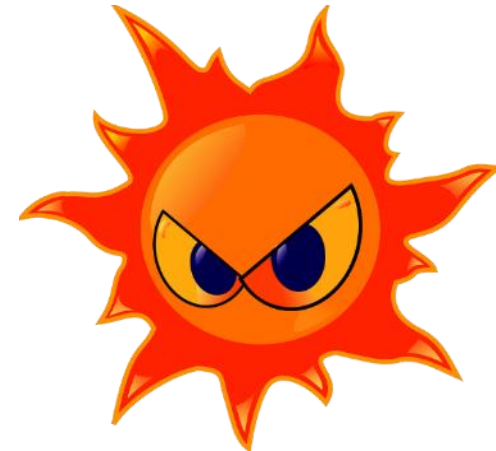


Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
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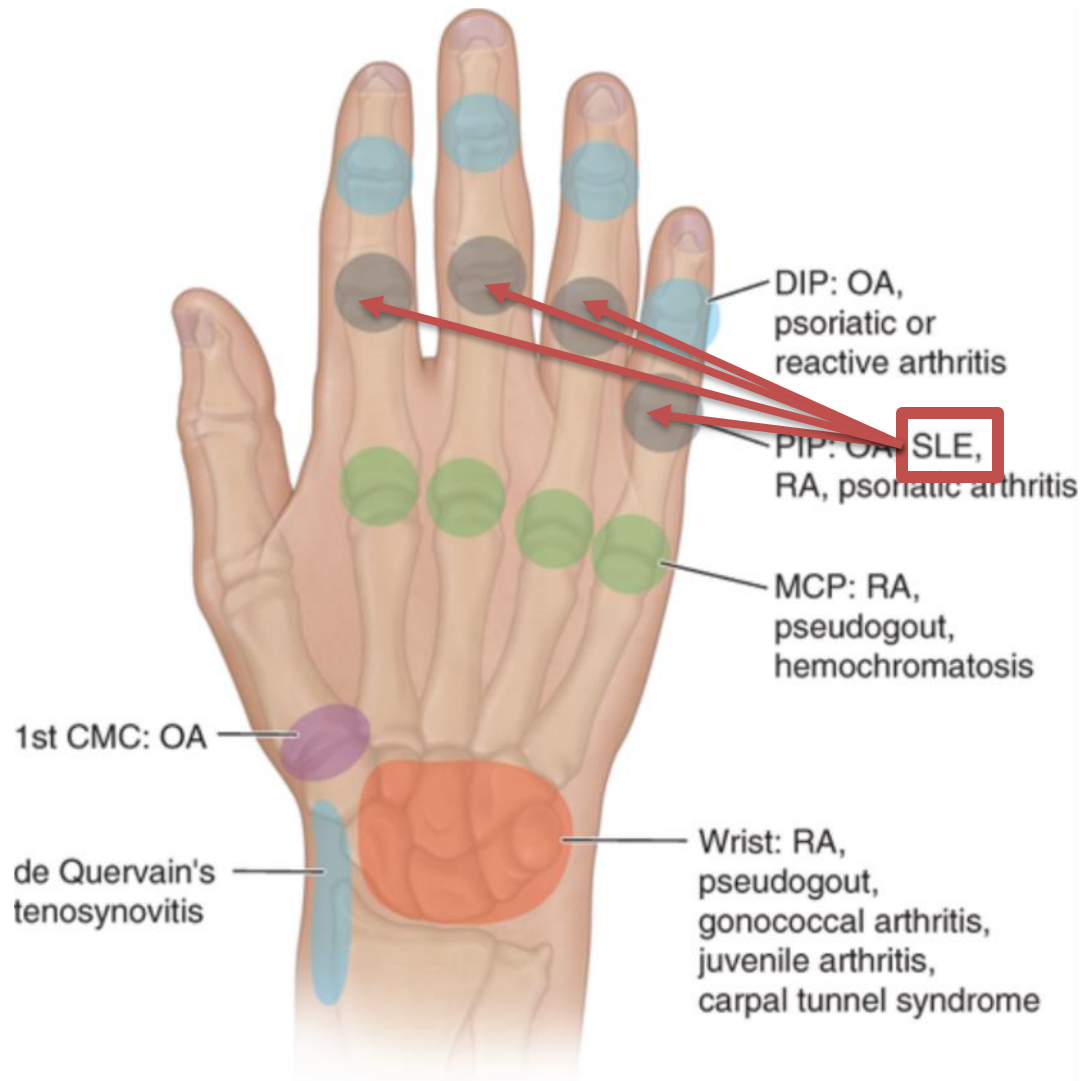
Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
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Discoid Rash of SLE



Source:
Discoid Lupus Erythematosus – MedScape
<https://emedicine.medscape.com/article/1065529-overview>

Joint involvement with SLE



Non-erosive
and spares the
DIP

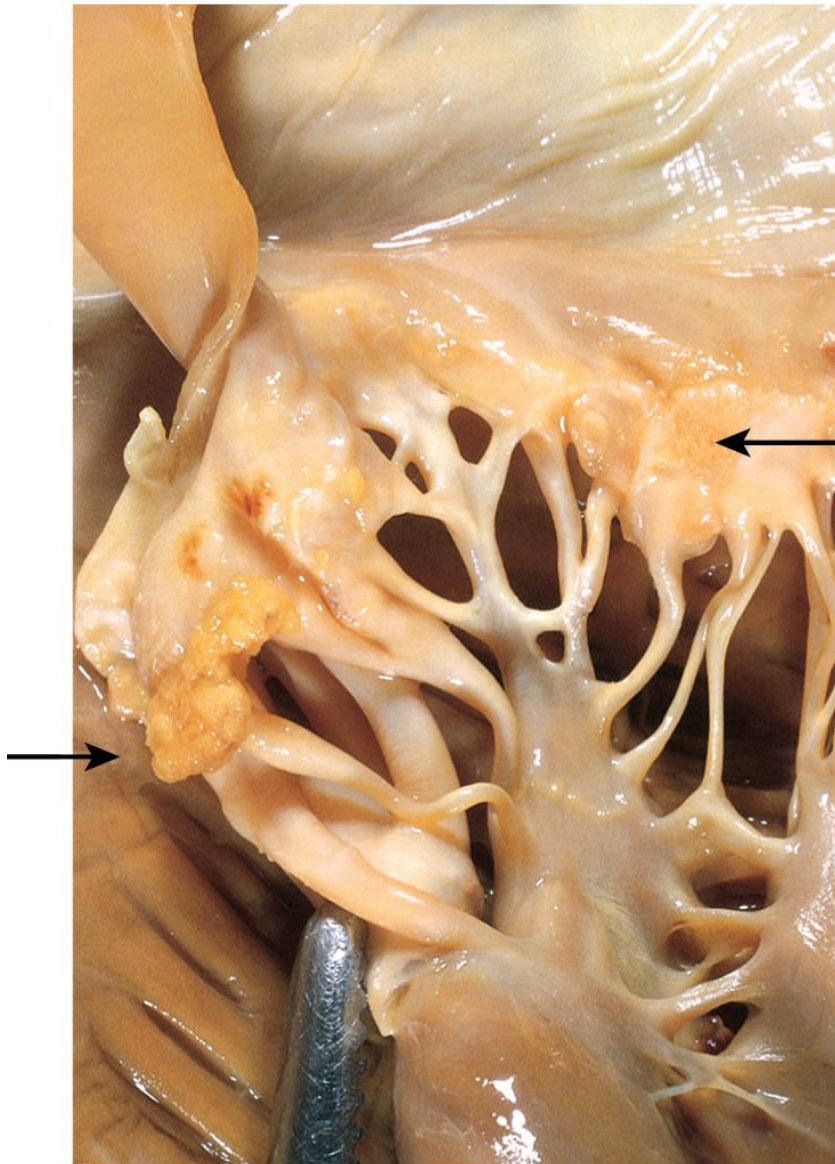
* No abnormal
finding on
XRAY*

Lung Involvement with SLE

- Pleuritis (pleurisy)
 - May be treated with NSAIDs or steroids
- Acute Lupus Pneumonitis
 - Requires high dose steroids and immunosuppressants
- Chronic (fibrotic) Lupus Pneumonitis
 - Permanent scarring
- Pulmonary Hypertension
 - Related to right heart failure



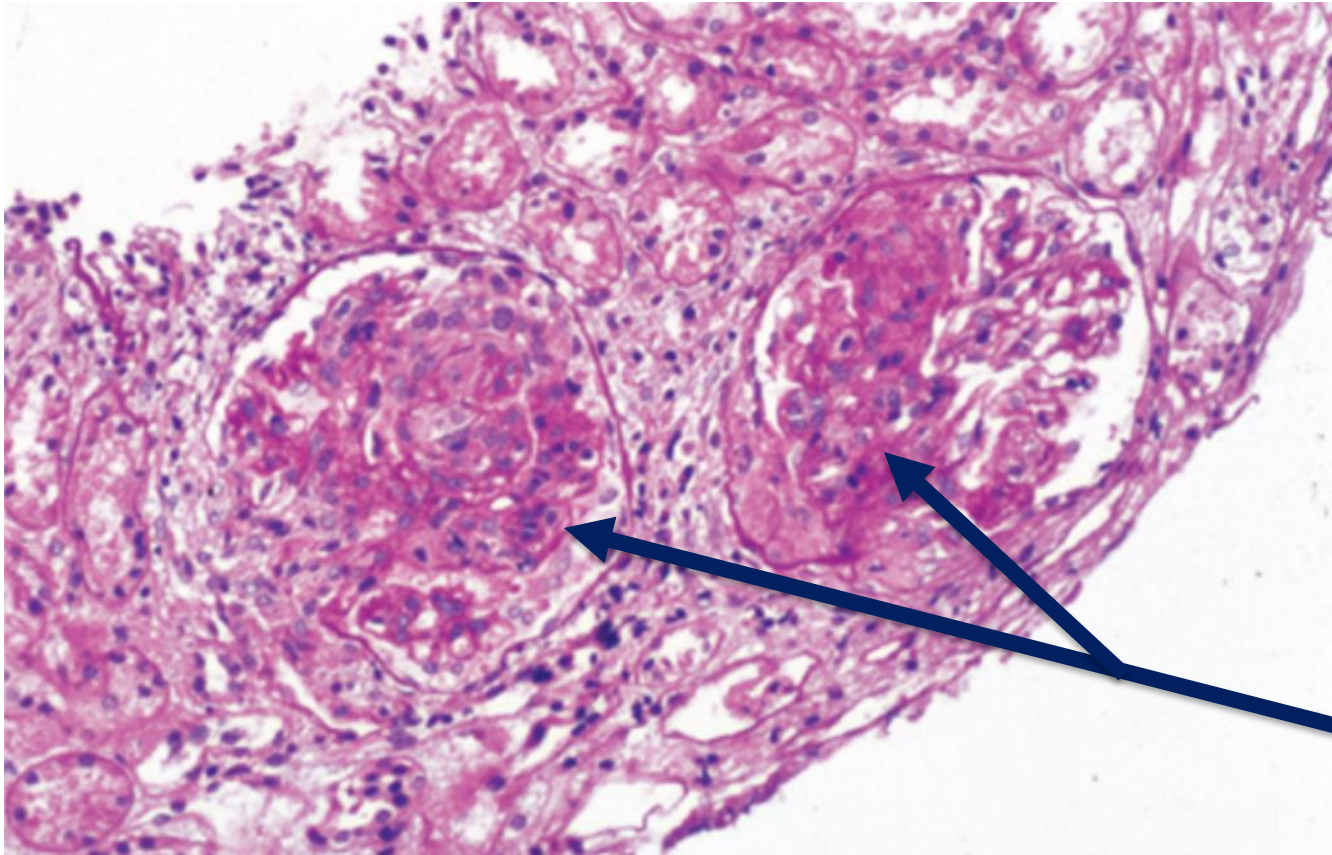
Cardiac effects related to SLE



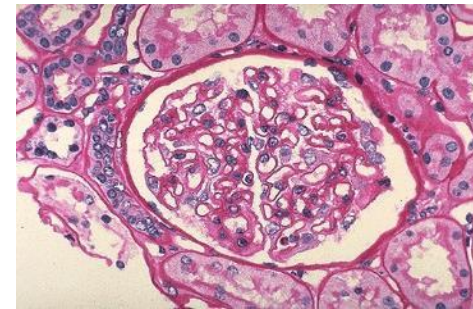
Libman-Sacks endocarditis of the mitral valve in lupus erythematosus. The vegetations attached to the margin of the thickened valve leaflet are indicated by arrows .

Jon C. Aster, MD, PhD; Abul K. Abbas, MBBS; Vinay Kumar, MBBS, MD, FRCPath; Jayanta Debnath, MD; Abhijit Das, MD, 6, 167-235
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Renal biopsy findings



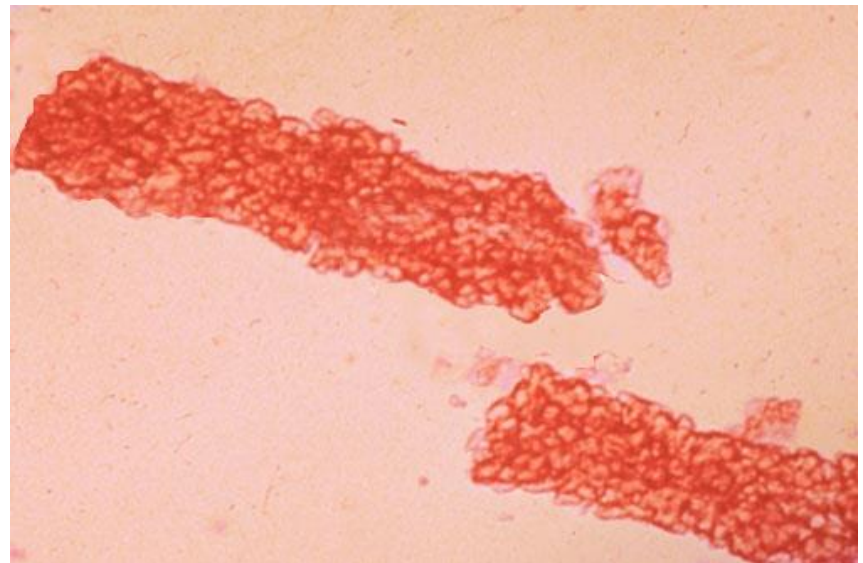
Endocapillary hypercellularity with segmental necrosis due to deposits (below is normal)



Citation: Atlas of Urinary Sediments and Renal Biopsies, Jameson J, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J. *Harrison's Principles of Internal Medicine*, 20e; 2018. Available at: <https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=196918516> Accessed: January 10, 2019

Renal Findings

- Proteinuria on urinalysis
- Red-cell casts on urine microscopy
- Endocapillary deposits of Ag-Ab complexes (next slide)



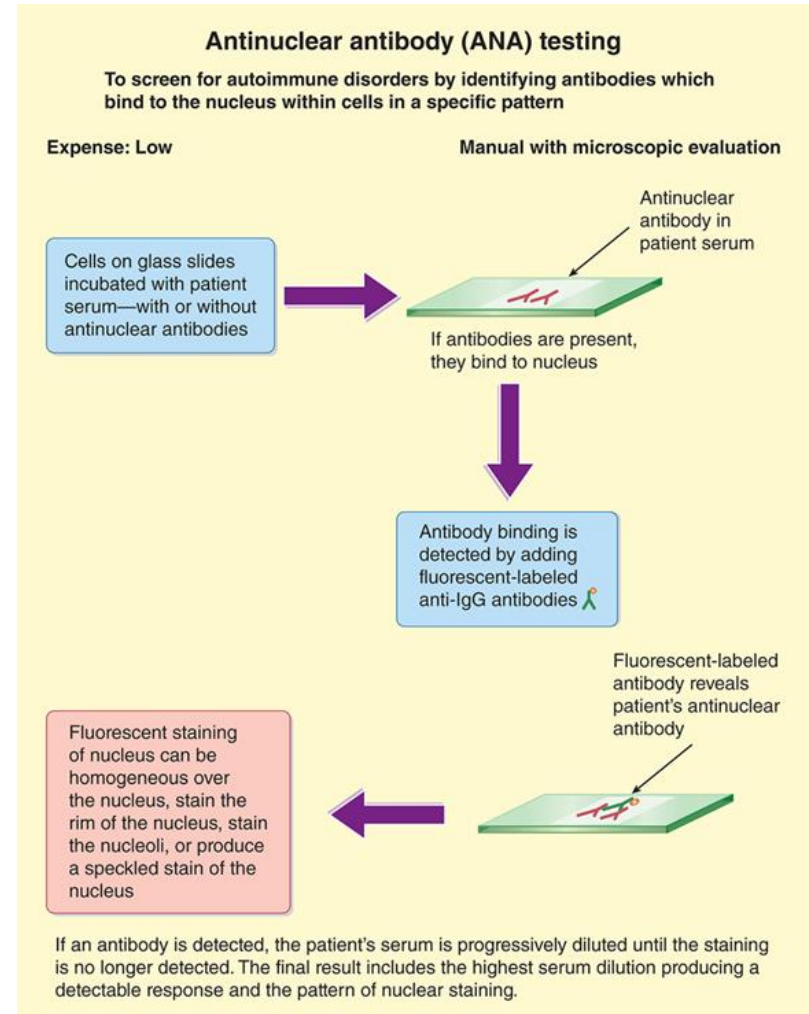
Source: Laboratory Info: <https://laboratoryinfo.com/types-of-casts-in-urine-and-their-clinical-significance/>

Neurological effects of SLE

- For neurologic effects to be diagnostic for SLE they must not be caused by medication or other disorders
 - Treatment of SLE can cause neurologic effects (ie chronic corticosteroid effects)
- SLE may effect:
- Central
 - HA, confusion, fatigue, depression, seizures
 - Peripheral
 - Peripheral neuropathy
 - Autonomic
 - Raynaud phenomenon
 - Livedo reticularis

Laboratory Testing

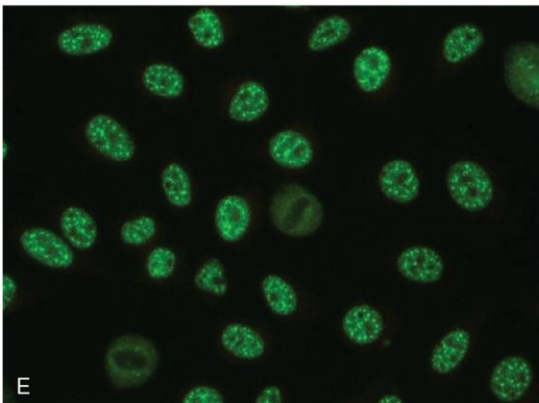
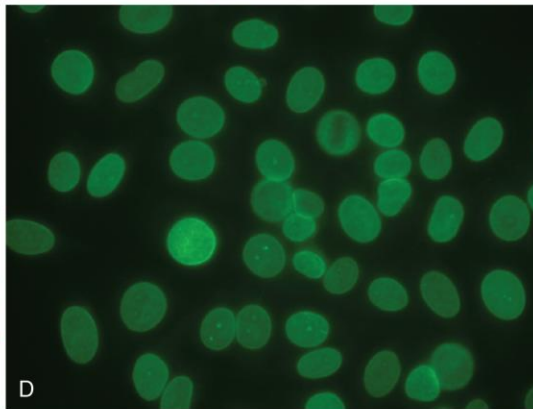
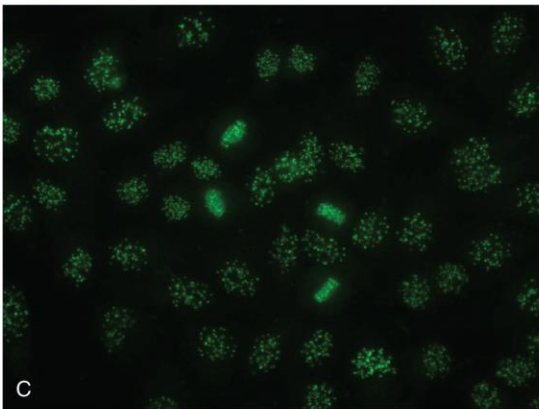
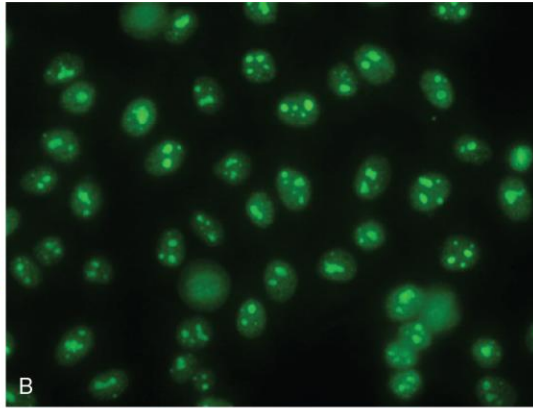
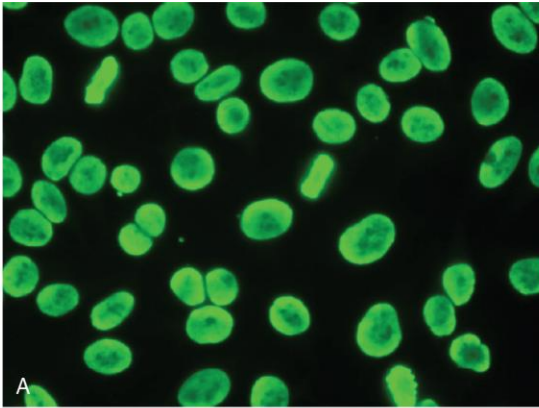
- Antinuclear Antibody
 - Low cost
 - Highly sensitive (~100%)
 - Variable specificity
 - If coupled with other tests then diagnostic reliability increased
 - **Anti-dsDNA Ab**
 - **Anti-Sm Ab**
 - Antiphospholipid Ab
 - Decreased complement



Source: Laposata's Laboratory Medicine, Chapter 2

<https://accessmedicine.mhmedical.com/content.aspx?sectionid=201361411&bookid=2503&jumpsectionID=201361423&Resultclick=2#1158019775>

Staining Patterns of Antinuclear antibodies



(A) Homogeneous staining of nuclei is typical of antibodies reactive with double-stranded DNA (dsDNA) and nucleosomes and is common in systemic lupus erythematosus (SLE). (B) Nucleolar clumpy pattern is typical of antibodies against the nucleolar protein fibrillarin (U3-RNP) and is strongly associated with systemic sclerosis. (C) Centromere pattern is seen predominantly in systemic sclerosis, primary biliary cirrhosis, and Sjögren syndrome. (D) Nuclear envelope pattern is caused by antibodies against nuclear membrane proteins, such as gp210, p68, and lamins, and can occur in patients with autoimmune liver diseases. (E) Coarse speckled pattern is seen with antibodies against Sm and U1-RNP, and can be observed in SLE, systemic sclerosis, and mixed connective tissue disease.

Jon C. Aster, MD, PhD; Abul K. Abbas, MBBS; Vinay Kumar, MBBS, MD, FRCPath; Jayanta Debnath, MD; Abhijit Das, MD, 6, 167-235

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ACR/EULAR Criteria for SLE

Entry criterion			
Antinuclear antibodies (ANA) at a titer of $\geq 1:80$ on HEp-2 cells or an equivalent positive test (ever)			
↓			
If absent, do not classify as SLE If present, apply additive criteria			
↓			
Additive criteria			
Do not count a criterion if there is a more likely explanation than SLE.			
Occurrence of a criterion on at least one occasion is sufficient.			
SLE classification requires at least one clinical criterion and ≥ 10 points.			
Criteria need not occur simultaneously.			
Within each domain, only the highest weighted criterion is counted toward the total score§.			
Clinical domains and criteria	Weight	Immunology domains and criteria	Weight
Constitutional		Antiphospholipid antibodies	
Fever	2	Anti-cardiolipin antibodies OR	
Hematologic		Anti- $\beta 2$ GP1 antibodies OR	
Leukopenia	3	Lupus anticoagulant	2
Thrombocytopenia	4	Complement proteins	
Autoimmune hemolysis	4	Low C3 OR low C4	3
Neuropsychiatric		Low C3 AND low C4	4
Delirium	2	SLE-specific antibodies	
Psychosis	3	Anti-dsDNA antibody* OR	
Seizure	5	Anti-Smith antibody	6
Mucocutaneous			
Non-scarring alopecia	2		
Oral ulcers	2		
Subacute cutaneous OR discoid lupus	4		
Acute cutaneous lupus	6		
Serosal			
Pleural or pericardial effusion	5		
Acute pericarditis	6		
Musculoskeletal			
Joint involvement	6		
Renal			
Proteinuria $>0.5\text{g}/24\text{h}$	4		
Renal biopsy Class II or V lupus nephritis	8		
Renal biopsy Class III or IV lupus nephritis	10		

Score 10 or more must be found for a diagnosis of SLE

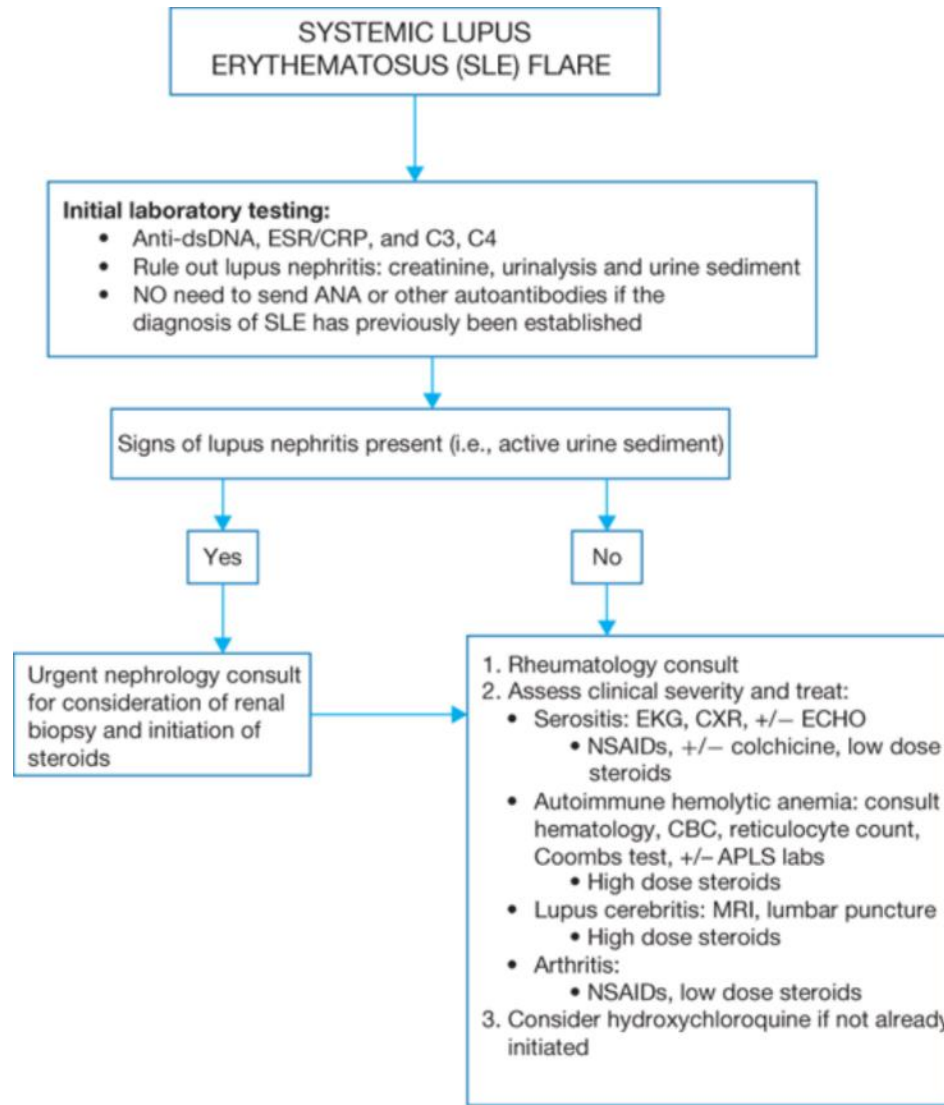
Source: <https://www.rheumatology.org/Portals/0/Files/1997%20Update%20of%201982%20Revised.pdf>

Pharmacologic Options for SLE Tx

- NSAIDS
 - Minor joint pain
- Hydroxychloroquine
 - Effective for lupus rashes and joint pain
 - Decreases incidence of severe disease
- Corticosteroids (topical)
 - Indicated for skin lesions
- Corticosteroids (PO/IV)
 - Required to control severe disease complications (end-organ inflammation and hematologic complications)
 - Start with high dose and taper to lowest dose that controls
- Immunosuppressant
 - Cyclophosphamide, Mycophenolate mofetil, Belimumab, Rituximab



Treatment plan for SLE Flare



Source: Navin Kumar, Anica Law: Teaching Rounds: A
Visual Aid to Teaching Internal Medicine Pearls on the Wards
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Non-pharmacologic Treatment Planning for SLE

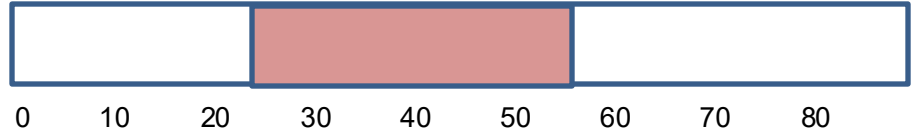
- Counselling and emotional support
- Multidisciplinary team approach
- Avoidance of UV light
- Generalized health promotion
- Screening for cardiovascular and renal disease
- Exercise and weight control
 - Osteoporosis
 - Medication side effects



Systemic Sclerosis: Epidemiology



Sex:
3:1 F:M



Onset at 25-55 years
Peak 50-60



Race/Ethnicity:
Earlier and more severe presentation in
African-American patients

Who

What is Systemic Sclerosis (Scleroderma)?

- Fibrosis of skin and internal organs
- Raynaud phenomenon is present in almost all patients with scleroderma
- Skin effects can be limited or diffuse
- Systemic effects: GI, Pulmonary, heart, muscles, and Renal

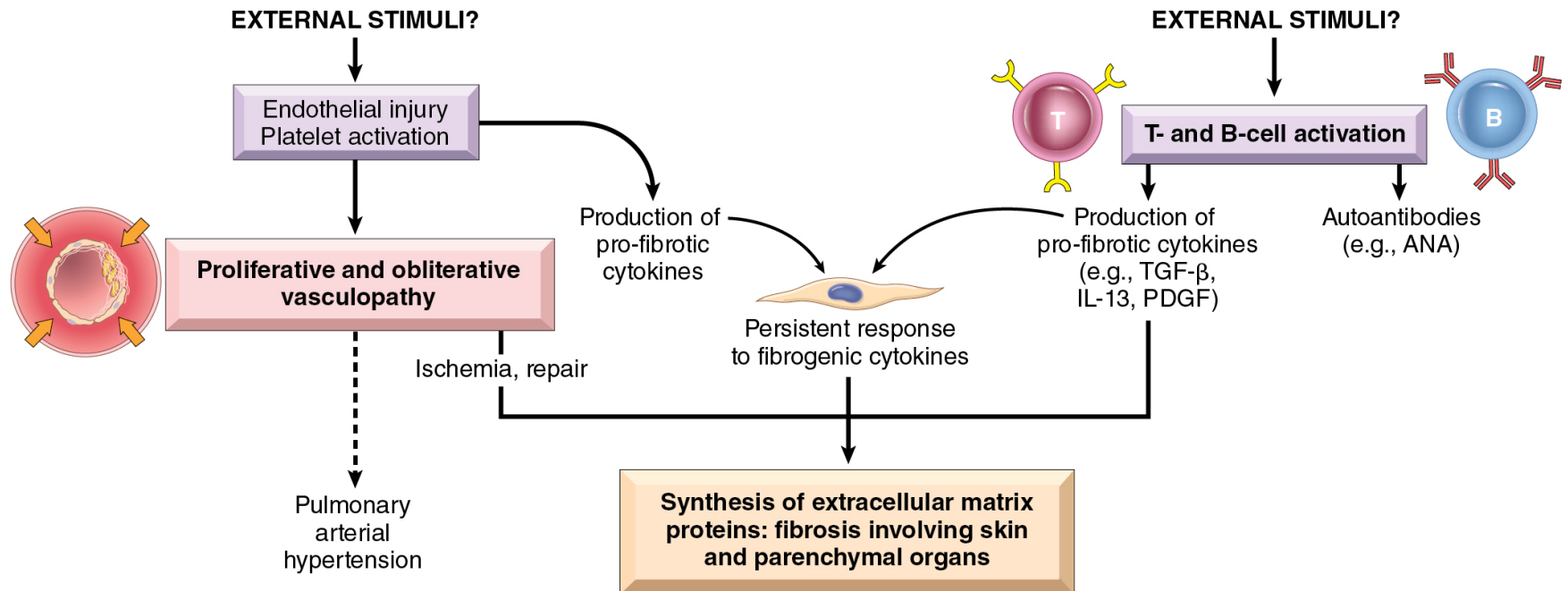
Raynaud's...

- Raynaud's phenomenon:
 - Numbness/tingling finger and toes
 - Episodic vasoconstriction of arteries/arterioles
 - Secondary to other condition
- Raynaud's disease
 - Primary / idiopathic
 - Younger women
 - milder



Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
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Pathogenesis of Systemic Sclerosis



- >>>Excess collagen



CREST Syndrome

- Calcinosis cutis
- Raynaud phenomenon
- Esophageal motility disorder
- Sclerodactyly
- Telangiectasia



Systemic Sclerosis (Scleroderma) findings

Calcinosis cutis



Source: R.P. Usatine, M.A. Smith, E.J. Mayeaux, Jr., H.S. Chumley
The Color Atlas and Synopsis of Family Medicine, Third Edition
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Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser,
D.L. Longo, J. Loscalzo: Harrison's Principles of Internal
Medicine, 20th Edition: www.accessmedicine.com
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Raynaud phenomenon



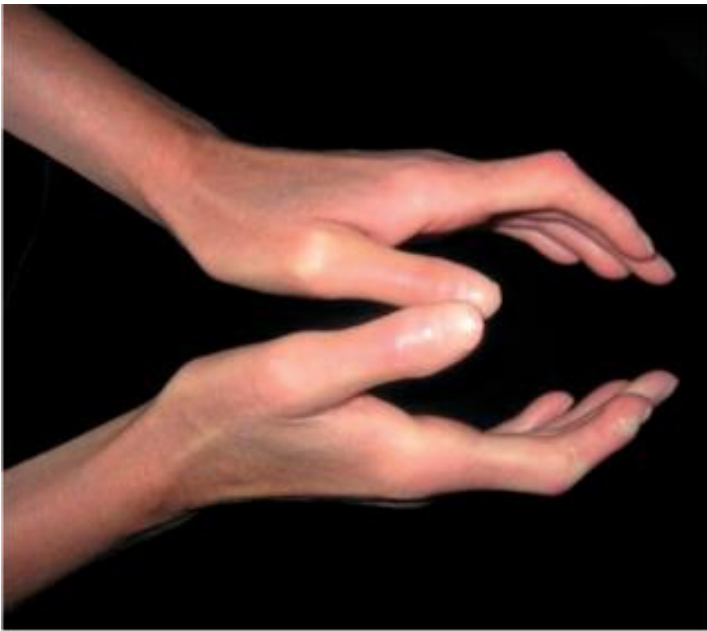
Source: K.J. Knoop, L.B. Stack, A.B. Storrow, R.J. Thurman:
The Atlas of Emergency Medicine, 4th Edition,
www.accessemergencymedicine.com
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Sources: as cited below each image
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Systemic Sclerosis (Scleroderma) findings

Scleryodactyly



A

Source: Sylvia C. McKean, John J. Ross, Daniel D. Dressler, Danielle B. Scheurer: Principles and Practice of Hospital Medicine, Second Edition, www.accessmedicine.com
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Telangiectasia



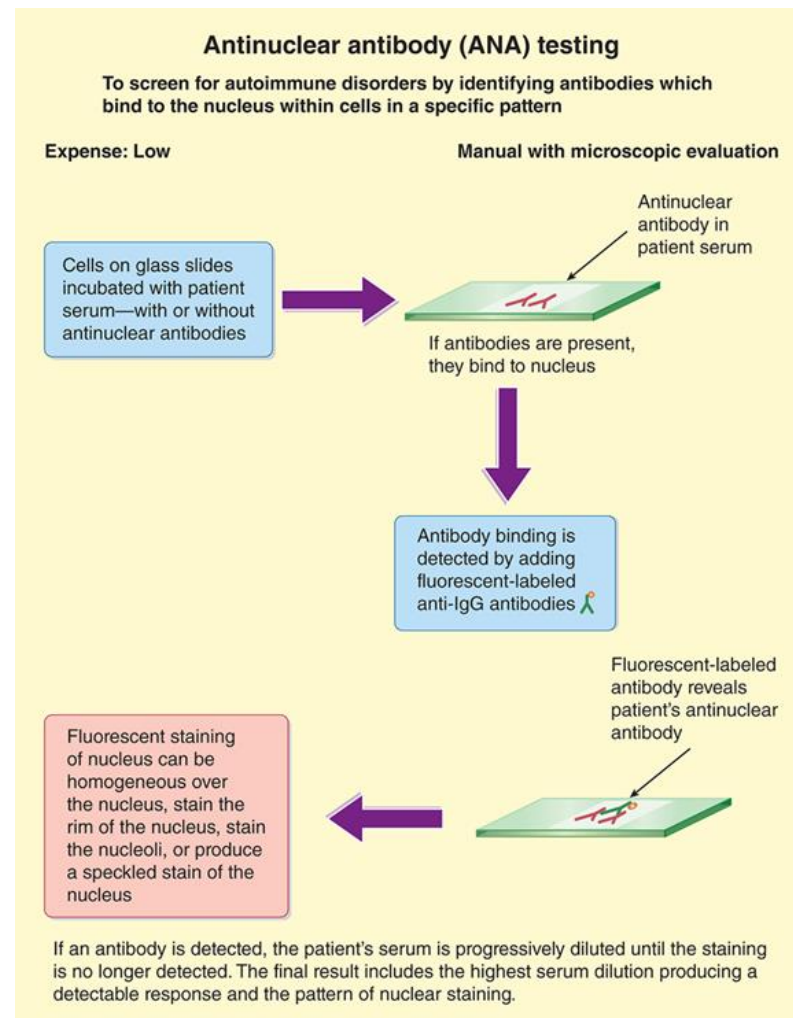
C

Source: Sylvia C. McKean, John J. Ross, Daniel D. Dressler, Danielle B. Scheurer: Principles and Practice of Hospital Medicine, Second Edition, www.accessmedicine.com
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Laboratory Testing for Systemic Sclerosis

- Antinuclear Antibody
 - Similar to SLE
 - Highly sensitive (~95%)
 - Improved specificity if coupled with:
 - **SCL-70** (scleroderma antibody)
 - Both forms
 - **Anti-centromere Ab**
 - Limited form
 - Anti-RNA polymerase III Ab
 - Related to diffuse form with renal disease



Treatment Plans of Systemic Sclerosis

Systems affected

- Raynauds
 - M/C treatment – Calcium channel blocker
- Esophageal Motility
 - Acid reduction (PPIs)
 - Oral prokinetic agents
 - Erythromycin
 - Metoclopramide
- Sclerodactyly
 - OT, surgical options

Diffuse treatments

- Methotrexate
- Cyclophosphamide
- IVIG
- Rituximab
- Tocilizumab
- Prednisone does NOT have a role in treatment: increases risk of renal crisis

Overview of treatment of Systemic Sclerosis

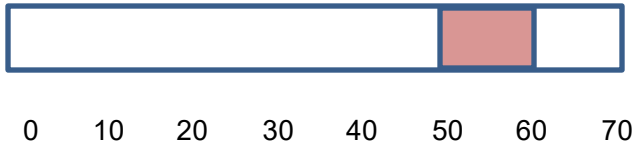
- Treatment is typically based on symptoms and organ systems effected
- Counselling needed for support throughout after diagnosis
- Nutritional support and guidance needed
- Discussion of prognosis encouraged to assist patients with understanding of disease progression.

Key points for Sjogren's Syndrome



Sex:
16:1 F:M

Who



Most common in females
between age of 50-60.
50% have another autoimmune
disorder

Diagnostic Criteria:

- Salivary gland inflammation by biopsy
- **Anti Ro/SSA or Anti La/SSB Abs**
- Oral signs or ocular signs of dry mouth or eyes for 3 months

Pathogenesis:

Gland: inflammatory infiltrates. Mix of polyclonal T and B cells early in course. Late in course: dominant B-Cell Clone.
5% develop lymphoma

Key targets

- Exocrine glands with decreased secretory function
 - Lacrimal glands
 - Salivary glands
- Can also effect lungs, thyroid, kidneys, skin and muscle

Source: Brandt JE, Priori R, Valesini G, Fairweather D. Sex differences in Sjögren's syndrome: a comprehensive review of immune mechanisms. *Biol Sex Differ*. 2015;6:19. Published 2015 Nov 3. doi:10.1186/s13293-015-0037-7

Key points for Sjogren's Syndrome

Key targets

- Exocrine glands with decreased secretory function
 - Lacrimal glands
 - Salivary glands
- Can also affect lungs, thyroid, kidneys, skin and muscle

Key Symptoms:

- Keratoconjunctivitis sicca: dry eyes
- Xerostomia: Dry mouth

Often associated with other autoimmune disorders:

- RA
- Raynaud's
- Thyroiditis

Treatment

- Cyclosporine ocular for dry eyes
- Pilocarpine for xerostomia
- Prednisone for severe systemic inflammation
- Avoidance of atropine like drugs

This is Sjögren's...

"I look healthy on the outside, however my disease is attacking my internal organs and destroying me from the inside out"

"Sjögren's has started to attack my lungs and I have difficulty breathing and performing everyday tasks"

"I rarely smile or talk in public since my inability to produce saliva has caused embarrassing tooth decay"

"I lost my passion for dining out because Sjögren's has affected my ability to chew and digest most foods"

"Joint pain, aching muscles and inability to sleep is an everyday occurrence for me"

"My inability to produce tears causes extreme eye discomfort and makes me prone to chronic eye infections"

"This terrible disease sets a 'new normal' way of life, which is 'not normal' at all"

"I wake up each day hoping a treatment is developed that will allow me to live an active and normal life again"

Sjögren's ("SHOW-grins") is a systemic autoimmune disease that affects the entire body. Along with symptoms of extensive dryness, other serious complications include profound fatigue, chronic pain, major organ involvement, neuropathies and lymphomas.

Neurological problems, concentration/memory-loss, dysautonomia

Dry nose, recurrent sinusitis, nose bleeds

Dry mouth, mouth sores, dental decay, difficulty with chewing, speech, taste and dentures

Fatigue, vasculitis, lymphoma, dry skin

Arthritis, muscle pain

Peripheral neuropathy, Raynaud's

Dry eyes, corneal ulcerations and infections

Swollen, painful parotid/salivary glands

Difficulty swallowing, heartburn, reflux, esophagitis

Recurrent bronchitis, interstitial lung disease, pneumonia

Abnormal liver function tests, chronic active autoimmune hepatitis, primary biliary cirrhosis

Stomach upset, gastroparesis, autoimmune pancreatitis

Irritable bowel

Interstitial cystitis

Vaginal dryness, vulvodynia

www.sjogrens.org



SSF16109.04

Overview of tests (SLE and SSc)

	RA	SLE	Sjögren	Diffuse Scleroderma	Limited scleroderma (CREST syndrome)	Polymyositis/ dermatomyositis
ANA	30–60	95–100	95	>95	>95	80
Anti-ds DNA	0–5	60	0	0	0	0
Rheumatoid Factor	70	20	75	30	30	33
Anti-Sm	0	10–25	0	0	0	0
Anti-SS-A	0–5	15–20	65	0	0	0
Anti-SS-B	0–2	5–20	65	0	0	0
Anti-SCL-70	0	0	0	33	20	0
Anti-Centromere	0	0	0	1	50	0
Anti-Jo-1	0	0	0	0	0	20–30
ANCA	0	0–1	0	0	0	0

Adapted from:
 Maxine A. Papadakis, Stephen J. McPhee, Jennifer Bernstein+ Table 20-7.
 Frequency (%) of autoantibodies in rheumatic diseases

Summary Slide

- Systemic Lupus Erythematosus, Systemic Sclerosis, and Sjogren's Syndrome are autoimmune disorders with multi-organ system challenges.
- Understanding of presenting findings and appropriate laboratory tests assist in the diagnosis.
- Treatment is multifactorial – both focusing on symptomatic resolution and limiting disease progression.

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>